Is Florida Getting Warmer?

Permutation Test on Key West Annual Mean Temperature (20th Century)

1. Methods

Annual mean temperatures recorded at Key West, Florida, were correlated with year. Let r_{obs} be the observed Pearson correlation coefficient:

$$r_{\text{obs}} = 0.5332$$

To test whether this correlation is greater than expected by chance, temperatures were randomly shuffled among years $N = 10000$ times, and the correlation recalculated each time:

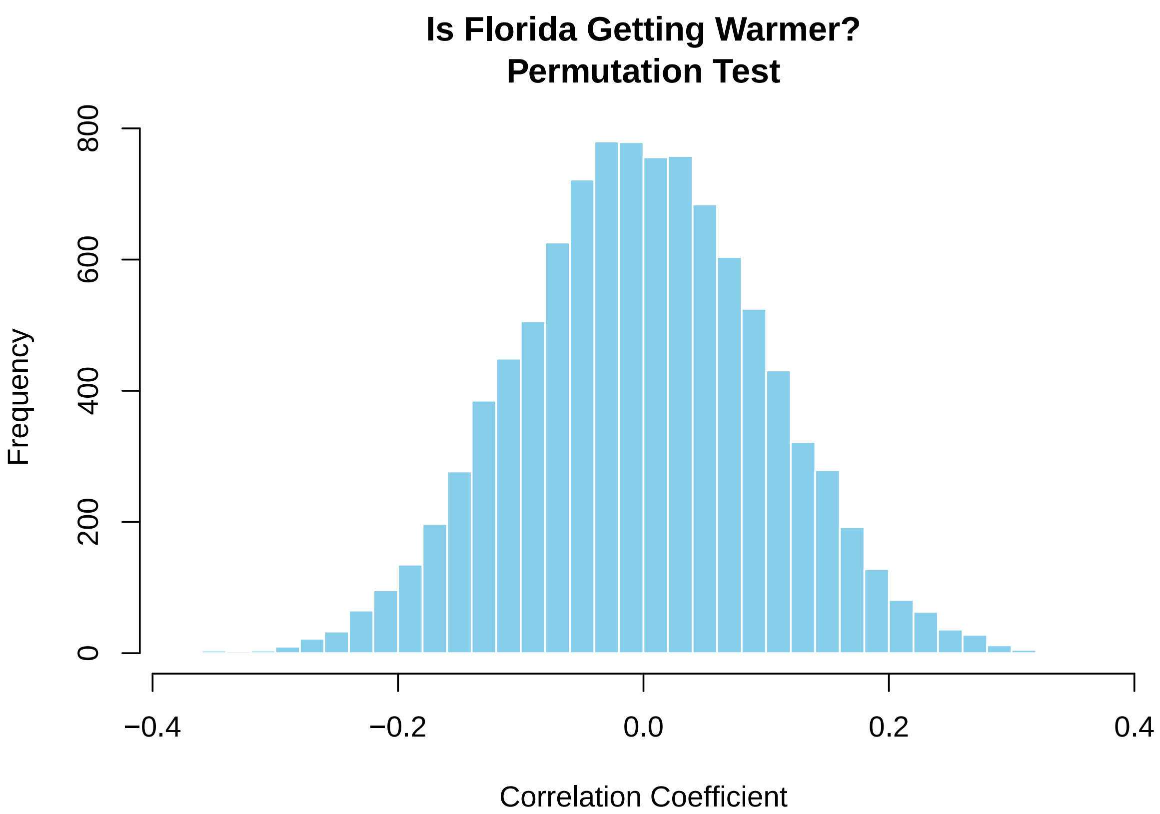
$$r_i = \text{cor}(\text{Year}, \text{sample}(\text{Temp}))$$

The empirical p-value is computed as the fraction of permuted correlations greater than or equal to the observed one:

$$p = \frac{\sum(r_i \geq r_{\text{obs}})}{N} = 0.0000$$

2. Results

The histogram below shows the distribution of correlation coefficients from the permutation test. The vertical red line marks the observed correlation r_{obs}.



Observed correlation: $r = 0.5332$
Empirical p-value: $p = 0.0000$
Number of permutations: $N = 10000$

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